Iziak Kwan

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TECHNICAL SKILLS

Engineering & Design: Engineering Design, System Integration, Circuit Analysis, Digital Logic, Troubleshooting, Experimental Testing, Data Analysis

Hardware: FPGA Development, Digital Logic, Electronic Circuits, Sensors and Measurement

Engineering Software: Quartus, ModelSim, Excel, GitHub, SolidWorks

Programming: C++, MATLAB

TECHNICAL PROJECTS

Digital Alarm Clock **May – July 2026**

Fundamentals of Digital Logic & Design (ENSC 252), SFU

- Designed and implemented a digital alarm clock using VHDL and FPGA hardware, integrating timekeeping, user input, display, and alarm functions.
- Developed a clock divider, time counter, finite state machine (FSM), button debouncer, seven-segment display driver, and alarm logic.
- Implemented operating modes for displaying time, setting the clock, and configuring the alarm using physical buttons.
- Simulated individual modules in ModelSim and integrated the complete system in Quartus to verify functionality and troubleshoot timing and display issues.

AM Radio Receiver **May – July 2026**

Electric Circuits 1 (ENSC 220), SFU

- Built an AM radio receiver using a breadboard, op-amp, diode, capacitors, resistors, and a custom-built inductor.
- Designed and tested the tuner, AM detector, and audio amplifier stages of the receiver.
- Used an oscilloscope and function generator to measure signals, analyze circuit behaviour, and troubleshoot electrical issues.
- Tested inductance, capacitance, and amplifier gain to improve station tuning and receiver performance.

Automotive Mechanical Repair & Troubleshooting

Independent Project | *Personal Vehicle Maintenance & Repair*

- Independently diagnosed and repaired mechanical issues involving the vehicle's suspension and drivetrain systems, including a displaced CV axle and seized ball joint.
- Applied hands-on mechanical troubleshooting techniques to remove seized components, realign drivetrain connections, and restore proper component fitment.
- Used automotive tools and mechanical principles to disassemble, inspect, repair, and reassemble vehicle components.
- Developed practical experience in mechanical systems, fault diagnosis, technical research, and systematic problem-solving through independent learning.

TECHNICAL PROJECTS CONTINUED

Electronic Dice Project **Oct 2024 – Nov 2024**

Electronics Laboratory Instruments Operation and Measurement Techniques (ENSC 120), SFU

- Designed and assembled an electronic dice circuit to generate randomized dice outcomes.
- Soldered electronic components and wiring onto a circuit board to construct a working prototype.
- Tested and troubleshot wiring, soldering, and circuit issues to verify functionality.

Automated Sidewalk Salter **Sept 2024 – Dec 2024**

Engineering, Science and Society (ENSC 100), SFU

- Designed and developed an automated sidewalk salting prototype to improve the consistency of salt distribution during winter conditions.
- Developed the mechanical dispensing mechanism and integrated sensors, motors, and control logic to automate salt application.
- Tested and refined the prototype to improve dispensing consistency, reliability, and overall functionality.

WORK EXPERIENCE

Barista **May 2026 – Present**

Gong Cha, Coquitlam, BC

- Follow standardized procedures to ensure accuracy, consistency, quality control, and compliance with food safety standards.
- Collaborate with team members to prioritize tasks, manage high-volume orders, and maintain efficient workflows under time constraints.
- Communicate effectively with customers and coworkers to clarify requirements, resolve issues, and prevent order errors.
- Demonstrate adaptability, attention to detail, and time management while balancing multiple tasks in a fast-paced environment.

Supervisor **Dec 2024 – Present**

Big Way Hot Pot, Coquitlam, Lougheed and Kingsway, BC

- Lead and delegate tasks based on team priorities and workload to improve workflow efficiency and maintain productivity during peak periods.
- Train and support employees on standard operating procedures, communicate expectations clearly, and promote consistent work quality.
- Apply problem-solving and decision-making skills to troubleshoot operational issues, resolve customer concerns, and coordinate solutions with management.
- Collaborate with employees and management across locations to communicate updates, coordinate responsibilities, and maintain consistent operational standards.

EDUCATION

BASc. in Engineering Science **Sept 2024 – Jun 2029**

Simon Fraser University, Burnaby, BC